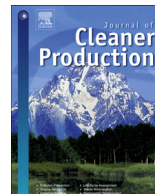




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journal homepage: www.elsevier.com/locate/jcleproRegional low-carbon economy efficiency in China: analysis based on the Super-SBM model with CO₂ emissionsJiarui Zhang ^a, Weihua Zeng ^{a,*}, Jinnan Wang ^b, Fengle Yang ^c, Hongqiang Jiang ^b^a School of Environment, Beijing Normal University, Beijing 100875, China^b Chinese Academy for Environmental Planning, Beijing 100012, China^c Yunnan Institute of Environmental Science, Kunming 650034, China

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ABSTRACT

As the largest energy consumer and CO₂ emitting country, the Chinese government is committing to a low-carbon economy. It is meaningful and contributes to evaluate and analyze the low-carbon economy efficiency (LCEE) of China. A Super-slack-based measure (Super-SBM) model with undesirable outputs, as combined with the Malmquist productivity index, is proposed to measure the LCEE and the dynamic low-carbon economy efficiency (DLCEE) of 30 provinces in mainland China from 2005 to 2012. The Theil index is also used to measure the rationalization level of industrial structure (RLIS) for discussing the improvements to the LCEE in China. The results indicate that the proposed undesirable outputs Super-SBM model can effectively rank the SBM-efficient provinces. China's regional economic development does not follow a low-carbon pattern, with an average LCEE of 0.517. As a whole, the China's economic development is gradually following a low-carbon development pattern with an annual LCEE improvement of 4.5%. The RLIS of China is also gradually changing better, as well. Through a comparative analysis of the LCEE and RLIS, the 30 provinces are divided into three sub-areas, and relevant suggestions are presented for improving the LCEE of different sub-areas in the future.

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1. Introduction

China has been the second largest economy in the world since 2010. However, because of high resource-consuming and low-efficiency economic development patterns, China became the world's largest energy consumer in 2010 (Du et al., 2011). It accounted for one-quarter of the world's total CO₂ emissions in 2011 (Choi et al., 2012), and devoured almost half of all coal produced in 2012 (Liu et al., 2013; Wang et al., 2013). According to the Stern Report, climate change will cause serious influence on the economic growth and social development, of which the associated loss and risk will be equivalent to 5%–20% of global GDP (gross domestic product) each year (Stern, 2006). To mitigate carbon emissions and climate change, China has been proposed a series of policies to prompt energy conservation and mitigate CO₂ emissions.

In the Chinese 12th Five-Year Plan (2011–2015), several new energy and carbon reduction targets were announced by 2015 with 2010 as the base year, such as reducing the energy intensity (energy consumption per unit of GDP) by 16% and decreasing the carbon intensity (CO₂ emissions per unit of GDP) by 17% (SCPRC, 2011). To achieve these goals, the best choice for China is to transform its economic development patterns and develop a low-carbon economy (LCE) (Liu et al., 2013; Wang and Chang, 2014).

The concept of LCE was first proposed in the British White Paper (DTI, 2003), which noted that the LCE was to obtain greater economic output through less natural resource consumption, leading to lower greenhouse gas emissions. Since the LCE was proposed, the policies related to LCE have been established in many countries, such as the United States, Germany, Japan and others (Shimada et al., 2007; Cherry et al., 2014). Dealing with climate change and developing the LCE have gradually become the consensus of the international community. The Chinese government has also committed to developing an LCE since the National Assessment Report of Climate Change was published in 2006. Until the present time, numerous studies have focused on energy conservation and emissions mitigation in China during the transition to an LCE. For

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instance, Jiang et al. (2010) analyzed China's energy development strategy under the LCE and proposed a series of energy saving measures and emissions reductions. Liu and Gallagher (2010) proposed to develop a possible carbon capture and storage (CCS) roadmap for an LCE in China. They found that coal gasification or polygeneration in combination with CCS could be a nearly unbeatable combination for China's low-carbon future. Liu et al. (2013) identified the major challenges of the LCE transition in China and proposed a five-pronged strategy to get China onto the low-carbon pathway. In addition, in an effort to develop an LCE, it is also meaningful in management to understand China's past low-carbon economy efficiency (LCEE), which is important for planning future development. As far as we know, almost no studies have evaluated and analyzed the LCEE of China. Due to the various socio-economic development levels in different regions, it is more meaningful to study the LCEE on a regional level rather than a national level (Chang et al., 2013). Thus, it is very necessary for us to quantitatively analyze the LCEE of different provinces in China, which will provide a realistic basis for developing an LCE in China.

Data Envelopment Analysis (DEA), a non-parametric mathematical programming approach, was first proposed to evaluate the relative efficiencies of decision making units (DMUs), i.e., the CCR-DEA model (Charnes et al., 1978). In recent years, DEA models have been widely applied to measure energy efficiency and CO₂ emission efficiency at the macro-economic level, because this method divides economic and environmental indexes into multiple inputs and outputs data (Guo et al., 2011). For instance, Lee and Lee (2009) analyzed the energy efficiency performance of 47 government office buildings in Taiwan using the DEA approach. Lu et al. (2013) measured the CO₂ emission efficiency in 32 OECD countries with a hybrid efficiency DEA model. Moreover, the DEA approach does not require functional form assumptions between inputs and outputs and can avoid man-made subjectivity for parameter weighting (Song et al., 2012). Therefore, the DEA method would more easily and scientifically provide an estimate of LCEE. Traditional DEA models (e.g., the CCR-DEA model) assume that all the outputs should be maximized for a given input level (Zhou et al., 2008). In fact, it is inevitable that undesirable outputs (e.g., CO₂ emissions) will be generated as by-products of desirable outputs (e.g., GDP) when using fossil fuels in the actual production process. According to the definition above, the basic idea of an LCE is to use fewer energy inputs (e.g., coal) to produce more desirable economic outputs while reducing undesirable greenhouse gas outputs (e.g., CO₂) as much as possible. In such cases, traditional DEA models are inappropriately applied to evaluate LCEE when considering CO₂ emissions.

The undesirable outputs slack-based measure (SBM) model was first proposed and extended to treat undesirable outputs by Tone (2001, 2004). It takes into account the slackness problems of inputs and outputs caused by the radial and angular choices (Song et al., 2012) and directly addresses the input excess and output shortfall in efficiency measurements. Therefore this method is particularly suitable for dealing with CO₂ emissions in the LCEE evaluation. In China, the undesirable outputs SBM model has been widely used to measure energy and environmental efficiency in recent studies. Choi et al. (2012) employed the undesirable outputs SBM model to estimate the potential reductions and efficiency of CO₂ emissions in China. They also used the model to estimate the marginal abatement costs of CO₂ emissions. Song et al. (2013a) calculated the environmental efficiency of 28 provinces in China from 1998 to 2009 and made a comparison with the traditional CCR-DEA model. The study proved that the undesirable outputs SBM model was more accurate and reliable than the CCR-DEA model.

However, the efficiency values obtained from the undesirable outputs SBM model are between 0 and 1. In most situations, there may be multiple SBM-efficient DMUs, i.e., the efficiency values may be simultaneously equal to 1. In such cases, the model cannot further rank the SBM-efficient DMUs. In actual management, it is needed to effectively rank the efficient DMUs. Therefore, this paper develops a Super-SBM model with undesirable outputs for evaluating the SBM-efficient DMUs, which has the high discriminating ability for further ranking the efficient DMUs. Then, the proposed undesirable outputs Super-SBM was used to evaluate the LCEE of 30 provinces in mainland China from 2005 to 2012. To track the LCEE performance changes over time, as combined with the Malmquist productivity index, the Dynamic low-carbon economy efficiency (DLCEE) was also computed for consecutive two-year periods during the study period. Furthermore, because China has been experiencing industrial transformations in recent years, the Theil index was used to measure the rationalization level of the industrial structure (RLIS) of the 30 provinces. Based on a comparative analysis of the LCEE and RLIS, relevant improvement measures were proposed for enhancing the provincial LCEE in China.

The rest of the paper is organized as follows. Section 2 describes the methodology in detail, which consists of the developed undesirable outputs Super-SBM model and the Malmquist productivity index. In Section 3, the data sources and descriptive statistics are represented. The results and an analysis of the empirical study are displayed in Section 4. Finally, a range of conclusions are presented in Section 5.

2. Methodology

2.1. Super-SBM model with undesirable outputs

Here, we present the Super-SBM model considering undesirable outputs based on the SBM DEA framework (Tone, 2001, 2002, 2004). We assume that technologies with more desirable outputs and less undesirable outputs relative to less resource inputs are considered efficient (Chang et al., 2013). We also assume that the LCE production system has n DMUs, and each unit has three factors: inputs, desirable outputs and undesirable outputs. Each unit uses m input factors to produce s_1 desirable outputs and s_2 undesirable outputs. If we denote the relative factors by three vectors: $x \in R^m$, $y^d \in R^{s_1}$ and $y^{ud} \in R^{s_2}$, the matrices X , Y^d and Y^{ud} can be defined as follows:

$$X = [x_1, x_2, \dots, x_n] \in R^{m \times n}, \quad Y^d = [y_1^d, y_2^d, \dots, y_n^d] \in R^{s_1 \times n}, \quad Y^{ud} = [y_1^{ud}, y_2^{ud}, \dots, y_n^{ud}] \in R^{s_2 \times n}.$$

Then, the production possibility set (PPS) can be described as follows:

$$P(x) = \left\{ (y^d, y^{ud}) \mid x \text{ produce } (y^d, y^{ud}), x \geq X\lambda, y^d \leq Y^d\lambda, y^{ud} \geq Y^{ud}\lambda, \lambda \geq 0 \right\} \quad (1)$$

where λ is the non-negative intensity vector, indicating that the above definition corresponds to the constant returns to scale (CRS) condition.

Based on this PPS, and according to Tone's SBM model (Tone, 2004), the SBM model dealing with undesirable outputs can be measured as follows:

$$\beta = \min \frac{1 - \frac{1}{m} \sum_{i=1}^m \frac{s_i^-}{x_{i0}}}{1 + \frac{1}{s_1 + s_2} \left(\sum_{r=1}^{s_1} \frac{s_r^d}{y_{r0}^d} + \sum_{t=1}^{s_2} \frac{s_t^{ud}}{y_{t0}^{ud}} \right)} \quad (2)$$

$$s.t. \quad x_0 = X\lambda + s^-$$

$$y_0^d = Y^d\lambda - s^d$$

$$y_0^{ud} = Y^{ud}\lambda + s^{ud}$$

$$s^- \geq 0, s^d \geq 0, s^{ud} \geq 0, \lambda \geq 0$$

where the vector s^d denotes the shortage of good outputs, and vectors s^- and s^{ud} correspond to excesses of inputs and undesirable outputs, respectively. The subscript 0 means a DMU whose efficiency is being estimated. The objective function β has values in the range [0, 1]. For a given evaluated DMU, if $\beta = 1$ and $s^- = s^d = s^{ud} = 0$, the DMU is SBM-efficient. If $\beta < 1$, the DMU is inefficient, and the inputs and outputs need to be improved.

The values obtained from Eq. (2) are between 0 and 1. To further rank the SBM-efficient DMUs, Tone proposed the Super-SBM model (Tone, 2002). However, the undesirable outputs are not considered in Tone's Super-SBM. Therefore, to more accurately evaluate efficiency, this paper proposes a Super-SBM model with undesirable outputs for evaluating the SBM-efficient DMUs.

Assume that the $DMU_k (x_k, y_k^d, y_k^{ud})$ is SBM-efficient, the corresponding Super-SBM model with undesirable outputs is given as follows:

$$\beta_{SE} = \min \frac{1 + \frac{1}{m} \sum_{i=1}^m \frac{s_i^-}{x_{ik}}}{1 - \frac{1}{s_1 + s_2} \left(\sum_{r=1}^{s_1} \frac{y_r^d}{y_{rk}^d} + \sum_{t=1}^{s_2} \frac{y_t^{ud}}{y_{tk}^{ud}} \right)}$$

$$s.t. \quad x_i^k \geq \sum_{j=1, j \neq k}^n x_{ij} \lambda_j - s_i^-$$

$$y_{rk}^d \leq \sum_{j=1, j \neq k}^n y_{rj}^d \lambda_j + y_r^d$$

$$y_{tk}^{ud} \geq \sum_{j=1, j \neq k}^n y_{tj}^{ud} \lambda_j + y_t^{ud}$$

$$1 - \frac{1}{s_1 + s_2} \left(\sum_{r=1}^{s_1} \frac{y_r^d}{y_{rk}^d} + \sum_{t=1}^{s_2} \frac{y_t^{ud}}{y_{tk}^{ud}} \right) > 0$$

$$\lambda, s^-, s^+ \geq 0$$

$$i = 1, 2, \dots, m; j = 1, 2, \dots, n (j \neq k) \quad r = 1, 2, \dots, s_1; \quad t = 1, 2, \dots, s_2 \quad (3)$$

where the β_{SE} is the objective function whose value can be more than 1, and all the other variables have the same meanings as in Eq. (2). This non-angle and non-radial Super-SBM model with undesirable outputs has the numerous advantages. First, it effectively solves the slackness problems of inputs and outputs caused by radial and angular choices (Song et al., 2012). Second, it directly addresses the input excess and output shortfall in efficiency measurements, which is particularly suitable for dealing with undesirable outputs (Zhou et al., 2006). Third, it can rank the

SBM-efficient DMUs in the SBM model (Song et al., 2013b). Therefore, compared with other DEA models, the Super-SBM model with undesirable outputs better represents the nature of regional LCE efficiency evaluations.

2.2. The Malmquist productivity index

The DEA models described in Eq. (2) and Eq. (3) are mainly used to evaluate cross-section LCEE comparisons between different DMUs. For a specific DMU, it is meaningful to track the changes in LCEE performance over time (Wu et al., 2012). The Malmquist productivity index (MPI) was first suggested by Malmquist (1953) as a quantity used in the analysis of inputs consumption. Färe et al. (1994) proposed the non-parametric DEA-based MPI for measuring productivity change over time. Since then, the nonparametric MPI has been widely applied in different areas (Wu et al., 2012; Huang et al., 2014). Although the MPI is defined based on a distance function, it can also be directly measured by the Super-SBM model with undesirable outputs. Assume that $\beta_t (x_j^t, y_j^{td}, y_j^{tud})$ and $\beta_s (x_j^s, y_j^{sd}, y_j^{sud})$ are the efficiency values of DMU_j based on its inputs, desirable outputs and undesirable outputs at period t for the reference technology at t and s . Further assume that $\beta_t (x_j^s, y_j^{sd}, y_j^{sud})$ and $\beta_s (x_j^t, y_j^{td}, y_j^{tud})$ are the efficiency values of DMU_j based on its inputs, desirable outputs and undesirable outputs at period s for the reference technology at t and s . The MPI can be defined as

$$MPI_j(t, s) = \left[\frac{\beta_t (x_j^s, y_j^{sd}, y_j^{sud}) \cdot \beta_s (x_j^t, y_j^{td}, y_j^{tud})}{\beta_t (x_j^t, y_j^{td}, y_j^{tud}) \cdot \beta_s (x_j^s, y_j^{sd}, y_j^{sud})} \right]^{1/2} \quad (4)$$

$MPI_{j(t, s)}$ can be used to measure the dynamic low-carbon economy efficiency (DLCEE) of DMU_j from period t to period s . $MPI_{j(t, s)} > 1$, $MPI_{j(t, s)} = 1$ and $MPI_{j(t, s)} < 1$ indicate that the LCEE of DMU_j has improved, remained unchanged or deteriorated from period t to s , respectively.

3. Data source and descriptive statistics

This study examined the LCEE of 30 provinces in mainland China (except Tibet, due to the absence of relevant energy data) from 2005 to 2012. Based on the principles of economics, labor and capital are typically the two basic production input elements at the macro-regional level (Chang et al., 2013). Energy consumption is also closely related to regional development (Berk and Yetkiner, 2014; Dhingra and Das, 2014). Thus, based on previous studies (Shi et al., 2010; Bian et al., 2015), we choose labor employment, capital stock and total energy consumption (TEC) as the input variables. As the main indicator for reflecting a region's economy (Song et al., 2013a), GDP is definitely selected as the desirable output variable. As the vital indicator characterizing the LCE, CO₂ emissions is selected as the undesirable output (Wang and Chang, 2014; Cherry et al., 2014). The labor employment variable refers to the persons employed at the end of each year in each province. TEC includes all types of energy such as petroleum, coal and natural gas, all of which are converted to tons of standard coal equivalent (TCE).

The labor employment, GDP and TEC data are collected from the China Statistical Yearbook (2006–2013), China Energy Statistical Yearbook (2013), and China Environment Statistical Yearbook (2006–2013), published by the National Bureau of Statistics of China. There is no official data available on provincial capital stocks and CO₂ emissions in China. In this study, the capital stock is

Table 1
CO₂ emissions factor by major carbonaceous fuel types in China.

Fuels	Coal	Coke	Kerosene	Petrol	Diesel	Fuel oil	Nature gas
CCF ^a	27.28	29.41	19.60	18.90	20.17	21.09	15.32
HE ^a	178.24	284.35	447.50	448.00	433.30	401.90	0.38
COF(%)	92.30	92.80	98.60	98.00	98.20	98.50	99.00

^a Notes: CCF and HE are expressed in units of tons carbon/trillion Joules and trillion Joules/10⁴ tons (m³), respectively.
Source: NDRC (2007) and Du et al. (2012).

calculated using the perpetual inventory method, which is expressed as follows (Zhang, 2008):

$$K_{it} = K_{it-1}(1 - \delta_{it}) + I_{it} \quad (5)$$

where for the i th province, K_{it} is the capital stock in year t , K_{it-1} is the capital stock in the previous year, δ_{it} is the capital depreciation rate in year t and I_{it} denotes the capital formation in year t . All the monetary variables in this study are converted into 2000 constant prices, in which GDP is deflated with GDP deflators, and capital stock is deflated using price indices of investment in fixed assets.

According to the IPCC (2006) guidelines for calculating CO₂ emissions, CO₂ emissions from burning fossil fuels can be estimated using the following equation (Du et al., 2012; Chang et al., 2013):

$$EC = \sum_{i=1}^7 E_i \times CEF_i = \sum_{i=1}^7 E_i \times CCF_i \times HE_i \times COF_i \times \left(\frac{44}{12} \right) \quad (6)$$

where EC denotes the quantity of CO₂ emissions from all seven types of fossil fuel consumption, i denotes the indicator of different fossil fuel types, including coal, coke, petrol, kerosene, diesel, fuel oil and natural gas, E_i denotes the total consumption of fossil fuel i , CEF_i is the CO₂ emission factor of a fuel, CCF_i is the carbon content factor (CCF) of fossil fuel i , HE_i is the heat equivalent (HE) of fossil fuel i , COF_i is the carbon oxidation factor (COF) of the carbonaceous fuel i and the number (44/12) denotes the ratio of the molecular weight of CO₂ (44) to the molecular weight of carbon (12). The Energy Research Institute (ERI) of the National Development and Reform Commission (NDRC) (2007) reported the CO₂ emission factors of major carbonaceous fuel types in China, as shown in Table 1.

Table 2 shows the descriptive statistics of the data. The mean labor employment value is 2579.66×10^4 persons, and that of capital stock is $12,215.69 \times 10^8$ Yuan (2000 constant prices). The TEC input is $11,616.65 \times 10^4$ TCE. The desirable output GDP is 9191.33×10^8 Yuan (2000 constant prices), whereas the undesirable output of CO₂ emissions is $26,509.48 \times 10^4$ tons.

Table 3 shows the correlation matrices for inputs and outputs, which clearly shows that all the correlation coefficients between the inputs and outputs are significantly positive. This indicates “isotonicity” of the inputs and outputs in the DEA model (Mostafa, 2009). Therefore, these data can be feasibly to be used for evaluating the provincial low-carbon economy efficiency.

Table 2
Descriptive statistics of input and output variables, 2005–2012, $N = 240$.

Inputs and outputs	Variable	Unit	Min	Max	Median	Mean	Std. dev.
Inputs	Labor employment	10 ⁴ persons	291.04	6554.30	2189.45	2579.66	1759.33
	Capital stock	10 ⁸ Yuan	1041.61	47,721.74	9871.31	12,215.69	9693.74
	TEC	10 ⁴ TCE	822.20	38,899.00	9111.18	11,616.65	7512.70
Desirable output	GDP	10 ⁸ Yuan	446.87	42,860.33	7133.05	9191.33	7839.21
Undesirable outputs	CO ₂ emissions	10 ⁴ tons	1069.97	89,854.88	19,689.26	25,427.13	18,673.80

4. Empirical analysis

4.1. Low-carbon economy efficiency analysis

Based on the National Western Development Strategy (NWDS), China's provinces are divided into three regions in this study, namely the east, central and west areas. The east area is comprised of eleven regions: three municipalities (Beijing, Tianjin and Shanghai) and eight coastal provinces (e.g., Shandong, Zhejiang and Guangdong). The central area is composed of eight inland provinces (e.g., Shanxi, Jilin, and Hunan). The west area includes one municipality (Chongqing) and ten provinces (e.g., Inner Mongolia, Guangxi and Sichuan). Using Eq. (2) and Eq. (3), we can estimate the LCEE of the 30 provinces in mainland China from 2005 to 2012, as shown in Table 4.

From Table 4, we can see that the LCEE values of Beijing, Shanghai and Guangdong are always greater than 1. Among these three provinces, Guangdong exhibits the largest value averaging 1.181, followed by Shanghai, which proves that the proposed undesirable outputs Super-SBM model can effectively rank the SBM-efficient provinces. Excluding Tianjin in 2012, the LCEE values of the other provinces are all less than 1. Overall, the average LCEE value in China during the study period is 0.517, which indicates that China's regional economic development dose not follow a low-carbon pattern during the study period. Table 4 also shows that the LCEE exhibits regional variation. In the eastern area, Guangdong has the highest average LCEE value while Hebei exhibits the lowest value of 0.398. In the central area, the LCEE values range from 0.562 to 0.306. In the west area, Chongqing registers for the highest LCEE value, while Ningxia is found to be the worst with the lowest value of 0.191.

Fig. 1 shows the LCEE during the study period. It can be found that the LCEE exhibits a slight upward trend among the three areas, which indicates that the LCEE has improved during the study period. Fig. 1 also shows that the LCEE of the west area gradually catch up with the level of central area, but the gap between east area and west/central area has become larger. The mean LCEE value of the east area is 0.755, followed by the central area, with a mean value of 0.432. The west area shows the lowest efficiency score of 0.332. This regional difference is found to be statistically significant using the non-parametric Kruskal–Wallis test, and the Chi-square value is 12.51 at a significance level of 1%. The results indicate that the low-carbon economic development level of the east area is more evident than the other two regions. In China, the economies of the east area provinces are relatively developed. The local governments in the east area can allocate more of their capital to develop low energy consumption industries, such as high-tech information industries, which rely on its strong economic and technical advantages. The provinces in the central area have undergone rapid regional development based on the regions abundant natural resources and strong industrial bases. However, because the industrial structure of the central area is not well managed, the economic growth has resulted in high energy consumption and increased CO₂ emissions. The economic development in the western area lags behind the eastern and central areas because of poorly

Table 3

Correlation matrices for inputs and outputs, 2005–2011.

	Labor employment	Capital stock	TEC	GDP	CO ₂ emissions
Labor employment	1				
Capital stock	0.639*	1			
TEC	0.753*	0.817*	1		
GDP	0.712*	0.923*	0.818*	1	
CO ₂ emissions	0.596*	0.653*	0.927*	0.635*	1

*Correlation is significant at the 1% level (2-tailed).

developed infrastructure. Although the socio-economic development levels have improved due to the NWDS, an enormous economic gap remains between the west provinces and the east/central area. To overcome this regional discrepancy, the central government or eastern and central provinces could provide a reasonable capital investment and technical support to the west areas (Zhang and Choi, 2013).

4.2. Dynamic low-carbon economy efficiency analysis

Using Eq. (4), we compute the DLCEE of the 30 provinces over time. Table 5 shows the DLCEE of 30 provinces for all of the consecutive two-year periods from 2005 to 2012.

Table 5 shows that during the study period, the 30 provinces experience an overall positive change ($=1.045$), indicating that the LCEE improved by 4.5% annually since 2005. All of the annual average DLCEE values display a positive shift (>1). The results indicate that Chinese economic development is turning into a low-carbon development pattern. Excluding Shanxi ($=0.999$), the provincial average DLCEE values all displays a positive shift (>1) during the study period, which implies that almost all of the provinces have an improvement in their LCEE. Among them, Tianjin, Jiangsu

and Hainan in the east area are found to have the highest annual average growth rate, with values greater than 7.0%. Based on the DLCEE analysis, we can see that China has made progress in converting to an LCE.

Fig. 2 shows the average DLCEE values of different areas over time. The DLCEE values of the east area are higher than the central and west areas from 2005/2006 to 2008/2009. However the DLCEE values of the west area are higher than the central and east areas from 2009/2010 to 2011/2012. The results indicate that the LCEE of the east area improves more than the other two areas during 2005–2009, while the west area improves the most from 2009 to 2012. The DLCEE values of the central area remain at a low level during the study period, which is probably because the central area has undertaken some energy-intensive and carbon-intensive industries transferred from the east area during the study years.

4.3. Discussions

China has long experienced traditional industrialization and industrial transformations, from the light/heavy industry to the technological intensive sector (Wang et al., 2013). China's economic growth mainly comes from the secondary industry (Fan et al.,

Table 4

The LCEE of 30 Provinces in China mainland, 2005–2012.

Province	2005	2006	2007	2008	2009	2010	2011	2012	Average
Beijing (E)	1.128	1.113	1.119	1.135	1.146	1.174	1.197	1.185	1.150
Tianjin (E)	0.708	0.713	0.693	0.728	0.724	0.723	0.743	1.035	0.758
Shanghai (E)	1.175	1.184	1.187	1.136	1.131	1.132	1.136	1.134	1.152
Hebei (E)	0.402	0.404	0.395	0.392	0.396	0.403	0.400	0.389	0.398
Liaoning (E)	0.581	0.579	0.578	0.572	0.592	0.601	0.593	0.584	0.585
Jiangsu (E)	0.666	0.674	0.679	0.693	0.723	0.720	0.695	0.671	0.690
Zhejiang (E)	0.652	0.645	0.629	0.626	0.636	0.639	0.621	0.620	0.634
Fujian (E)	0.706	0.700	0.677	0.678	0.690	0.693	0.652	0.675	0.684
Shandong (E)	0.577	0.571	0.576	0.578	0.595	0.605	0.610	0.613	0.591
Guangdong (E)	1.197	1.182	1.186	1.203	1.184	1.176	1.168	1.154	1.181
Hainan (E)	0.623	0.591	0.576	0.554	0.554	0.574	0.558	0.669	0.587
Anhui (C)	0.438	0.435	0.431	0.417	0.429	0.447	0.454	0.477	0.441
Jiangxi (C)	0.438	0.434	0.432	0.439	0.461	0.466	0.468	0.483	0.453
Heilongjiang (C)	0.560	0.566	0.556	0.553	0.566	0.566	0.577	0.550	0.562
Henan (C)	0.387	0.379	0.369	0.360	0.363	0.368	0.364	0.373	0.370
Hubei (C)	0.367	0.365	0.365	0.379	0.393	0.414	0.422	0.433	0.392
Hunan (C)	0.442	0.440	0.441	0.454	0.466	0.485	0.485	0.449	0.458
Jilin (C)	0.470	0.467	0.463	0.459	0.475	0.485	0.482	0.483	0.473
Shanxi (C)	0.315	0.307	0.311	0.305	0.294	0.300	0.303	0.311	0.306
Gansu (W)	0.317	0.316	0.314	0.306	0.319	0.320	0.325	0.334	0.319
Guangxi (W)	0.396	0.393	0.385	0.383	0.381	0.387	0.385	0.405	0.389
Guizhou (W)	0.214	0.213	0.218	0.226	0.240	0.259	0.273	0.287	0.241
Inner Mongolia (W)	0.395	0.386	0.393	0.402	0.401	0.401	0.393	0.406	0.397
Ningxia (W)	0.185	0.183	0.183	0.188	0.187	0.194	0.188	0.219	0.191
Qinghai (W)	0.239	0.239	0.237	0.245	0.253	0.271	0.273	0.276	0.254
Shaanxi (W)	0.340	0.336	0.338	0.331	0.357	0.367	0.375	0.389	0.354
Sichuan (W)	0.391	0.362	0.355	0.339	0.351	0.376	0.424	0.425	0.378
Chongqing (W)	0.430	0.422	0.420	0.424	0.446	0.469	0.481	0.479	0.446
Xinjiang (W)	0.359	0.347	0.343	0.343	0.341	0.340	0.339	0.326	0.342
Yunnan (W)	0.352	0.339	0.331	0.327	0.334	0.335	0.341	0.354	0.339
Annual average	0.515	0.510	0.506	0.506	0.514	0.523	0.524	0.540	0.517

Note: E, C and W in parentheses, refer to the east, central and west areas, respectively.

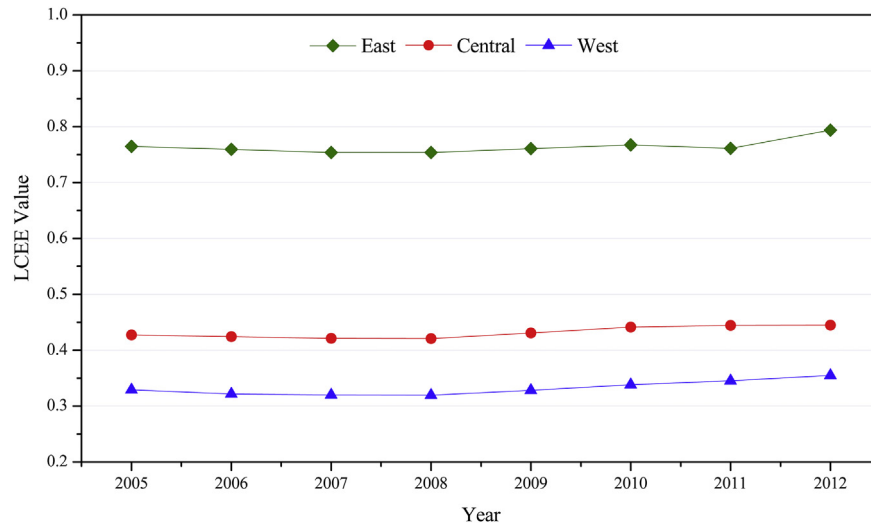


Fig. 1. The average LCEE values of different regions over time.

2007). Among the three industries, the second industry uses much more energy and generates more CO₂ than the other two industries. The proportion of the GDP linked to the secondary industry in 2012 was 48.7%, while the industry sector accounted for 40.6% of the GDP. The TEC of the industry sector was 24.64×10^4 TCE, accounting for 70.82% of the TEC of China. A reasonable industrial structure can promote effective resource utilization and economic growth (Guan et al., 2008). Therefore, an appropriate industrial structure is vital for China's shift to an LCE.

The quality of relationships between different industries can be used to represent the rationalization of industrial structure, which

includes two major aspects. The first is reflecting the coordination degree among different industries. The second is reflecting the effective utilization level of resources in different industries. The Theil index, first proposed by Theil (Theil, 1967), is used to measure the differences between samples. In recent years, it has been widely applied to measure regional income and energy intensity inequalities (Gustafsson, 2002; Alcantara and Duro, 2004). The advantage of Theil index is that it can measure the contribution of the gaps between intra-group and inter-group to the total gap, thereby avoiding the calculation of absolute values. Therefore, this paper applied the Theil index to measure the rationalization levels

Table 5
The DLCEE values of 30 provinces from 2005/2006 to 2011/2012.

Province	2005/2006	2006/2007	2007/2008	2008/2009	2009/2010	2010/2011	2011/2012	Average
Beijin (E)	1.021	1.045	1.031	1.033	1.034	1.062	1.075	1.043
Tianjin (E)	1.074	1.055	1.073	1.064	1.059	1.091	1.074	1.070
Shanghai (E)	1.043	1.053	0.978	1.028	1.031	1.031	1.032	1.028
Hebei (E)	1.059	1.039	1.038	1.020	1.048	1.029	1.008	1.034
Liaoning (E)	1.068	1.073	1.047	1.065	1.057	1.027	1.044	1.054
Jiangsu (E)	1.083	1.089	1.094	1.088	1.047	1.057	1.048	1.072
Zhejiang (E)	1.060	1.051	1.067	1.049	1.049	1.027	1.094	1.057
Fujian (E)	1.062	1.042	1.073	1.055	1.052	0.995	1.064	1.049
Shandong (E)	1.040	1.073	1.057	1.052	1.050	1.067	1.058	1.057
Guangdong (E)	1.037	1.079	1.039	1.010	1.022	0.998	1.052	1.034
Hainan (E)	1.075	1.071	1.076	1.067	1.076	1.061	1.071	1.071
Anhui (C)	1.029	1.034	1.010	1.027	1.056	1.037	1.083	1.039
Jiangxi (C)	1.032	1.047	1.074	1.063	1.027	1.038	1.084	1.052
Heilongjiang (C)	1.062	1.040	1.044	1.036	1.027	1.055	1.003	1.038
Henan (C)	1.020	1.021	1.021	1.015	1.035	1.022	1.059	1.028
Hubei (C)	1.039	1.054	1.087	1.047	1.078	1.052	1.075	1.062
Hunan (C)	1.037	1.053	1.073	1.034	1.060	1.029	0.769	1.008
Jilin (C)	1.045	1.053	1.043	1.053	1.055	1.035	1.058	1.049
Shanxi (C)	0.996	1.003	1.011	0.987	0.995	0.996	1.005	0.999
Gansu (W)	1.038	1.040	1.006	1.039	1.022	1.035	1.054	1.033
Guangxi (W)	1.035	1.033	1.049	1.009	1.033	1.028	1.105	1.042
Guizhou (W)	1.022	1.058	1.061	1.049	1.092	1.069	1.060	1.059
Inner Mongolia (W)	1.037	1.079	1.039	1.010	1.022	1.041	1.053	1.021
Ningxia (W)	1.046	1.056	1.072	1.004	1.073	1.005	1.105	1.052
Qinghai (W)	1.057	1.053	1.080	1.045	1.109	1.042	1.066	1.065
Shaanxi (W)	1.032	1.058	1.027	1.084	1.052	1.056	1.063	1.053
Sichuan (W)	0.969	1.030	1.000	1.040	1.089	1.103	1.076	1.052
Chongqing (W)	1.035	1.059	1.070	1.074	1.082	1.068	1.064	1.065
Xinjiang (W)	1.031	1.054	1.057	1.010	1.034	1.037	0.999	1.032
Yunnan (W)	0.995	1.017	1.026	1.013	1.020	1.038	1.058	1.027
Annual average	1.040	1.050	1.046	1.038	1.051	1.041	1.049	1.045

Note: E, C and W, in parentheses, refer to the east, central and west areas, respectively.

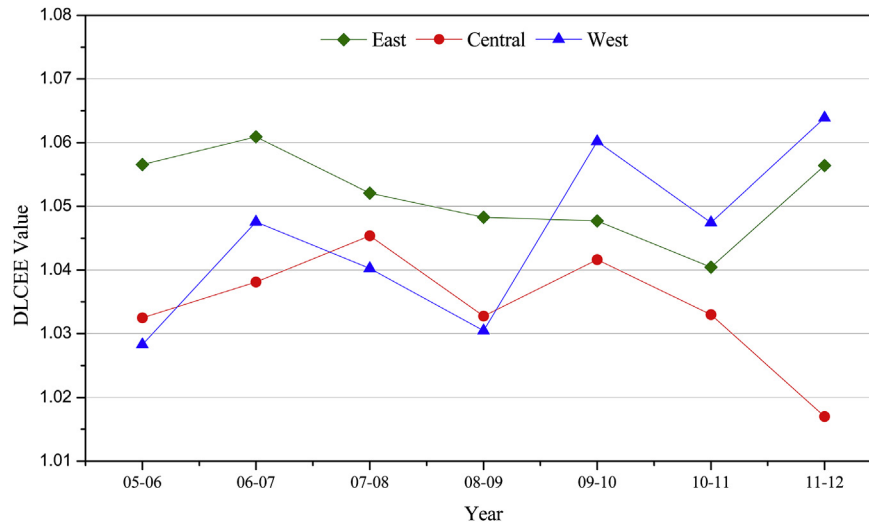


Fig. 2. The average DLCEE values of different areas over time.

of the industrial structures (RLIS) of the 30 provinces, which can be expressed as follows:

$$TL_i = \sum_{j=1}^3 \left(Y_{ij}/Y \right) \ln \frac{Y_{ij}/Y}{L_{ij}/L} \quad (7)$$

where TL_i is the RLIS of the i th province, and a higher value of TL indicates a larger industrial structure deviation degree, which also means that the industrial structure tends to be unreasonable, Y_{ij} stands for the j th industry value ($j = 1, 2, 3$) in the i th province, Y_i

stands for the GDP of i th province, L_{ij} denotes the number of employed persons of the j th industry in the i th province and L_i denotes the number of the employed persons in the i th province.

Table 6 shows the RLIS values of 30 provinces from 2005 to 2012. It can be seen from Table 6 that the China's annual average RLIS values have been improved from 0.285 in 2005 to 0.276 in 2012, which indicates that the RLIS of China is gradually improving. Table 6 also shows that the RLIS varies across different areas. In the east area, Beijing has the highest average RLIS value, while Shandong exhibits the lowest value of 0.215. In the central area, the average RLIS values range from 0.358 to 0.162. In the west area,

Table 6

The RLIS of 30 provinces in mainland China, 2005–2012.

Province	2005	2006	2007	2008	2009	2010	2011	2012	Average
Beijing (E)	0.039	0.037	0.037	0.037	0.038	0.039	0.033	0.031	0.036
Tianjin (E)	0.092	0.099	0.084	0.086	0.078	0.069	0.069	0.064	0.080
Shanghai (E)	0.128	0.143	0.126	0.147	0.132	0.016	0.016	0.025	0.092
Hebei (E)	0.222	0.222	0.192	0.197	0.179	0.171	0.167	0.147	0.187
Liaoning (E)	0.167	0.182	0.175	0.199	0.193	0.207	0.215	0.200	0.192
Jiangsu (E)	0.169	0.155	0.130	0.120	0.109	0.099	0.089	0.082	0.119
Zhejiang (E)	0.113	0.106	0.091	0.086	0.081	0.065	0.055	0.056	0.081
Fujian (E)	0.162	0.151	0.132	0.119	0.117	0.115	0.098	0.089	0.123
Shandong (E)	0.246	0.245	0.220	0.228	0.216	0.202	0.186	0.174	0.215
Guangdong (E)	0.200	0.189	0.188	0.179	0.177	0.162	0.135	0.134	0.171
Hainan (E)	0.154	0.189	0.193	0.170	0.161	0.156	0.151	0.147	0.165
Anhui (C)	0.219	0.221	0.192	0.180	0.188	0.218	0.242	0.223	0.210
Jiangxi (C)	0.138	0.161	0.163	0.160	0.156	0.179	0.179	0.158	0.162
Heilongjiang (C)	0.353	0.353	0.303	0.311	0.269	0.290	0.274	0.233	0.298
Henan (C)	0.331	0.344	0.309	0.294	0.266	0.253	0.245	0.229	0.284
Hubei (C)	0.244	0.266	0.267	0.261	0.294	0.302	0.308	0.302	0.280
Hunan (C)	0.229	0.217	0.192	0.198	0.202	0.214	0.223	0.222	0.212
Jilin (C)	0.238	0.257	0.276	0.284	0.289	0.333	0.340	0.323	0.292
Shanxi (C)	0.383	0.371	0.381	0.402	0.324	0.342	0.351	0.311	0.358
Gansu (W)	0.506	0.536	0.539	0.523	0.507	0.520	0.529	0.501	0.520
Guangxi (W)	0.314	0.331	0.297	0.317	0.305	0.331	0.327	0.340	0.320
Guizhou (W)	0.824	0.768	0.777	0.734	0.702	0.662	0.641	0.596	0.713
Inner Mongolia (W)	0.394	0.447	0.447	0.464	0.479	0.486	0.482	0.458	0.457
Ningxia (W)	0.320	0.304	0.322	0.299	0.257	0.463	0.481	0.472	0.365
Qinghai (W)	0.399	0.397	0.359	0.372	0.345	0.338	0.346	0.323	0.360
Shaanxi (W)	0.541	0.438	0.561	0.518	0.454	0.476	0.423	0.966	0.547
Sichuan (W)	0.231	0.234	0.200	0.207	0.221	0.236	0.239	0.227	0.224
Chongqing (W)	0.296	0.358	0.359	0.366	0.350	0.345	0.311	0.264	0.331
Xinjiang (W)	0.312	0.408	0.371	0.417	0.336	0.343	0.360	0.330	0.360
Yunnan (W)	0.597	0.587	0.558	0.521	0.490	0.512	0.463	0.426	0.519
Annual average	0.285	0.290	0.281	0.280	0.264	0.272	0.266	0.268	0.276

Note: E, C and W, in parentheses, refer to the east, central and west areas, respectively.

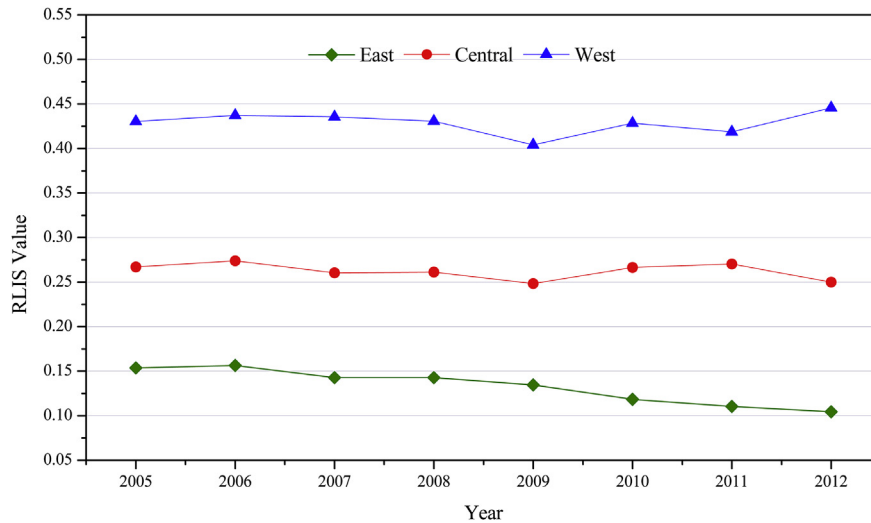


Fig. 3. The average RLIS values of different areas over time.

Sichuan registers the highest average RLIS value of 0.224, while Guizhou exhibits the lowest value of 0.713, which was also the lowest during the study period.

Fig. 3 shows the average RLIS values of different areas over time. It can be seen that the RLIS of east area is always higher than the other two areas, followed by the central area. The average east area RLIS values show a declining trend while, while RLIS values in the other areas fluctuate during the study period. The results indicate

that it is more needed for the central and west area to adjust and optimize their industry structure when get onto the LCE pathway.

The average RLIS and LCEE values of each province are illustrated in Fig. 4. As shown in Fig. 4, the 30 provinces can be divided into three sub-areas. Sub-area one contains seven east area provinces (e.g., Beijing, Shanghai, Zhejiang, and Fujian), which are the most developed provinces in China. Sub-area two contains fourteen provinces, including four east area provinces (Liaoning, Hebei,

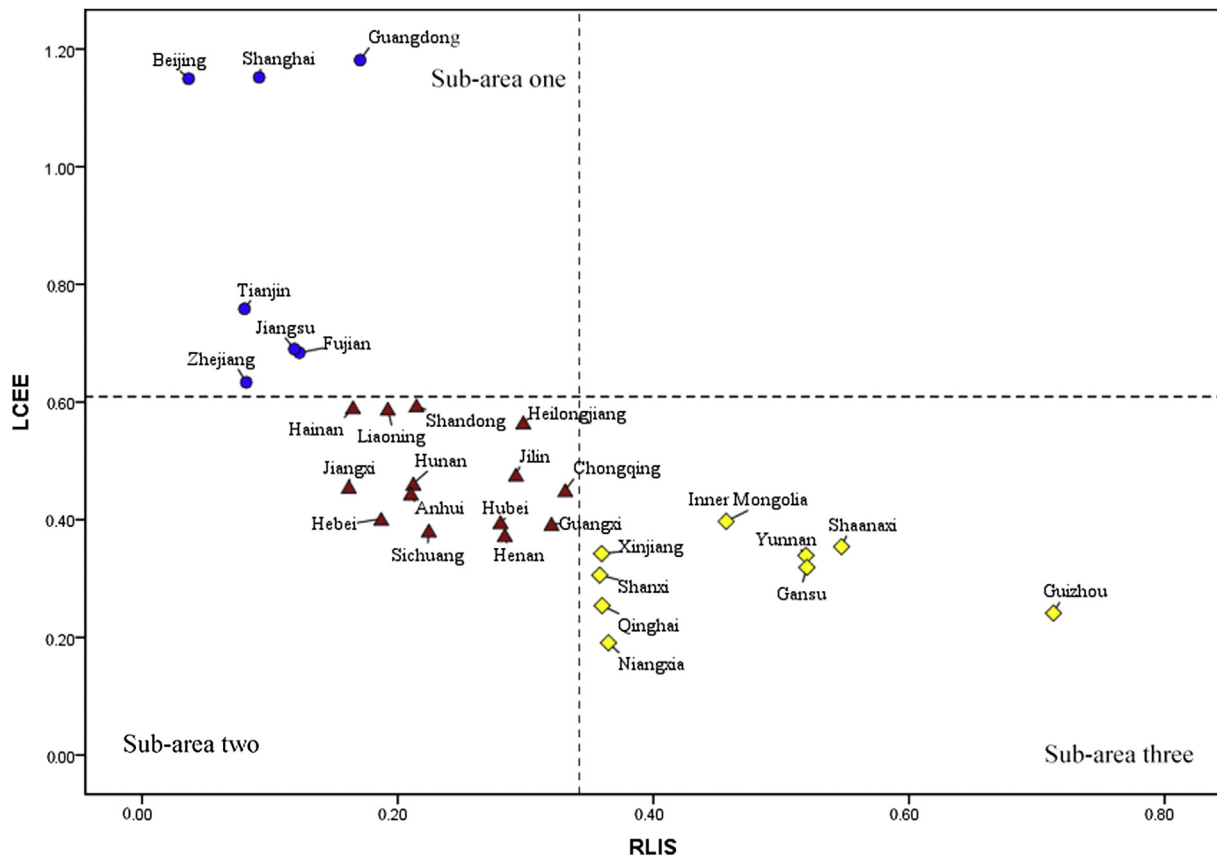


Fig. 4. Average RLIS and LCEE values of each province.

Shandong and Hainan), seven central provinces (e.g., Anhui, Jiangxi, and Hunan) and three west area provinces (Guangxi, Sichuan and Chongqing). Sub-area three includes nine provinces, which are all west area provinces except Shanxi.

The average LCEE and RLIS values of Sub-area one are 0.893 and 0.100, respectively. The industrial structure in these provinces is better than the other two sub-areas. The technological sectors are not energy-intensive with lower energy dependences and fewer CO₂ emissions (Lee and Hashim, 2014). Thus, relying on the strong economic and technical advantages, these provinces should commit to converting high energy consumption industries to technological industries, such as high-tech information sectors, to further promote an LCE.

The average RLIS value of Sub-area two is 0.241. Sub-area two accounted for 48.98% of the total GDP and 51.17% of the total CO₂ emissions of China in 2012. There are many heavy industry bases in Sub-area two, including Heilongjiang, Hebei and Shandong. Heavy industries are energy-intensive (e.g., iron and steel industries) and have the largest CO₂ emissions of all secondary industries (Du et al., 2011). To improve the LCEE, these heavy industry bases should encourage and support technical renovations in traditional heavy industries and eliminate backward production techniques and equipment. Through these measures, heavy industries can improve their energy utilization efficiencies and mitigate CO₂ emissions (Guan et al., 2008; Liu and Gallagher, 2010). In addition, because it borders Sub-area one, Sub-area two should take advantage of its nature and labor resources to attract investments or new technologies from the developed provinces in Sub-area one.

Shanxi and Inner Mongolia accounted for 36.15% GDP and 49.88% of CO₂ emissions from Sub-area three in 2012. They are the major coal producing provinces in China, and their economic growth is excessively dependent on the coal industry. These two provinces should upgrade and improve the coal industry, as well as promote economic diversification on the way to an LCE (Liu et al., 2013). Given the weak economic foundation in Sub-area three, those provinces should focus on improving their socio-economic levels. A regional coordinated development mechanism should be established to improve their LCEE. The central government or the provinces in Sub-area one and Sub-area two should offer external financial and technical support for these provinces.

5. Conclusions

A Super-SBM model with undesirable outputs is established to evaluate the LCEE of 30 provinces in mainland China from 2005 to 2012. Combined the Malmquist productivity index, the DLCEE is also computed for all consecutive two-year periods. The results show that the LCEE values of Beijing, Shanghai and Guangdong are always greater than 1, among which Guangdong is the largest, with an average value of 1.181. This proves that the proposed undesirable outputs Super-SBM model can effectively rank the SBM-efficient provinces. Overall, China's regional economic development dose not a low-carbon pattern, with an average LCEE value of 0.517. There are distinct differences in the LCEE among the east, central and west areas. As a whole, the LCEE has improved by 4.5% annually since 2005, and all of the provinces, except Shanxi, have improved in their LCEEs. These results indicate that Chinese economic development is converting to a low-carbon development model.

The Theil index is also applied to measure the RLIS values of 30 provinces in China. The results show that the RLIS varies across different areas and the RLIS of China is gradually improving. China's annual average RLIS values have improved from 0.285 to 0.276. During the study period, the RLIS value of east area is always better than the other two areas and exhibited a declining trend, while the RLIS values of the other two areas exists fluctuated. To further

develop an LCE, adjust and optimize the industry structure is more needed for the central and west area.

Through a comparative analysis of the LCEE and RLIS values, the researched 30 provinces are divided into three sub-areas. Sub-area one contains seven east area provinces, Sub-area two contains four east area provinces, seven central provinces and three west area provinces, and Sub-area three includes all west area provinces except Shanxi. To improve LCEE and develop an LCE, different suggestions are given for the three sub-areas. (1) The socio-economic of Sub-area one is the most developed in China, with the high level of LCEE and RLIS levels. The provinces in Sub-area one should commit to developing technologically intensive industries. (2) Sub-area two includes numerous heavy industry bases, which should encourage and support technical renovations in traditional heavy industries, eliminate backward production techniques and equipment for mitigating CO₂ emissions. (3) A regional coordinated development mechanism should be established to provide external financial and technical support to the provinces in Sub-area three.

However, some points should be mentioned. First of all, due to the absence of official provincial capital stock and CO₂ emissions data, we have to calculate values based on previous studies (Zhang, 2008; Du et al., 2012; Chang et al., 2013). Therefore, the data may cause biased estimation on the results, which is a common weakness of empirical studies. Furthermore, China's LCEE is only compared among the Chinese provinces included in this study. If compared to other advanced countries (e.g., OECD countries), the efficiency values would have likely been much worse. Based on data availability, this study could also be conducted using the same method to find the LCEE ranks of China and other advanced countries in the future, which can help to further improve the LCEE of China.

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